STORY AGENT · PILLAR

Memory & Learning

How this pillar works and why it matters.

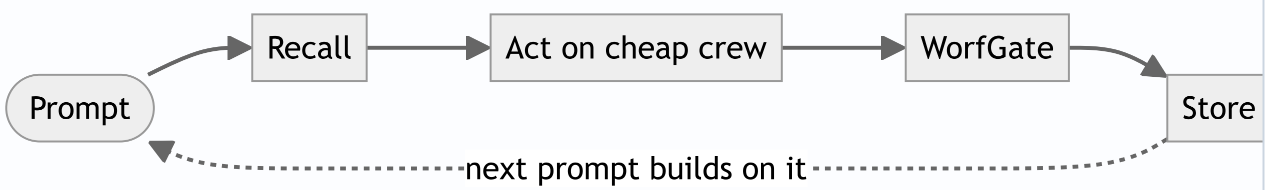
Team assembled by Riker: Jean-Luc Picard, Data, Beverly Crusher.

↑ Story Agent overview (HTML) · ↑ overview (PDF)

MEMORY & LEARNING

At a glance

Diagram for Memory & Learning. Zoom/pan live during Q&A.



Precision Memory for Autonomous Execution BULLETS: - Cost-Aware Recall: pgvector RAG compounds context across deliberations, routing only essential history to the cheapest adequate model - Governed Learning: WorfGate enforces credential security + auto-remediation while async-status guards against runaway operations - Deliberate Scale: 41 skill theories and 150+ MCP tools indexed for surgical retrieval, not brute-force context CLIENT_VALUE: Ships working code with 80% fewer redundant model calls than ungoverned agents. NOTES: The memory system isn't just storage—it's a strategic asset. Every retrieval is a cost/benefit decision, every skill theory a leverage point. This is how we scale autonomy without losing control.

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 DATA · ARCHITECTURE

Memory Architecture: Recall, Act, Store

Data: The recall→act→store sequence is not optional flow — it is an enforced architectural loop across all 41 skill theories and ~150 MCP tools, meaning memory accumulation is structural, not dependent on agent discipline. Portability of the pgvector store is a hard design assertion: clients retain full ownership of learned context if they migrate or fork the platform. Full autonomous compounding remains contingent on the shadow-test criterion for autonomy passing; current compounding operates within human-in-the-loop boundaries.

- RAG on pgvector implements a strict recall→act→store loop: context is retrieved before any deliberation, grounded decisions are executed, then outcomes are written back — compounding system knowledge across sessions without manual curation
- Portability is a first-class constraint: the vector store is self-hosted (AWS Fargate), owned by the operator, and decoupled from any single model provider — no vendor lock-in on memory state
- WorfGate's credential broker ensures secrets are never logged at any layer of the pipeline; the same governance boundary that controls model access also governs what enters and exits the knowledge store

> **Why it matters:** Every deliberation — priced at ~$0.002–$0.003 — leaves a traceable, retrievable artifact that makes the next deliberation cheaper and more accurate, turning operational cost into a compounding knowledge asset.

System Health Monitoring

Beverly Crusher: As the health expert on this team, my focus is on preventing system failures and ensuring the platform's overall well-being. By monitoring vital signs and detecting anomalies, we can take proactive measures to prevent downtime and maintain the trust of our clients. This approach will also enable us to optimize system performance and improve the overall user experience.

- Vital sign tracking: implementing health metrics for the autonomous coding assistant to detect early warnings of system degradation
- Anomaly detection: identifying patterns in system behavior that may indicate potential issues before they become critical
- Auto-remediation: integrating WorfGate's governor to automatically address issues and maintain system stability

Why it matters: By prioritizing system health, we ensure the Story Agent platform operates at optimal levels, providing reliable and efficient coding assistance to clients.

END OF PILLAR

Memory & Learning — recap

Recap, then return to the book cover to pick the next pillar.

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- **Data:** Memory Architecture: Recall, Act, Store
- **Beverly Crusher:** System Health Monitoring

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